

Project Title: Hybrid QML and Quantum String Algorithms for Efficient Document Extraction and Processing

Duration: 6 Weeks

Intern Level: Undergraduate/Graduate with Quantum Software & ML background

Mentor: Potluri Krishna Priyatham

Objective

To design and simulate a resource-efficient pipeline for document intelligence that replaces oversized classical Large Language Models (LLMs). The project utilizes Quantum Machine Learning (QML) for computer vision-based text extraction from images/PDFs and Quantum String Algorithms for targeted data manipulation, ensuring high precision with reduced computational overhead.

Week-by-Week Plan

Week 1: Vision Foundation & Data Encoding

Goal: Establish the quantum representation for document imagery.

- Study Flexible Representation for Quantum Images (FRQI) and Novel Efficient Quantum Representation (NEQR).
- Implement pixel-to-quantum state encoding for PDF-derived grayscale images.
- Deliverable: Technical summary of encoding efficiency vs. classical normalization.

Week 2: QML for Feature Extraction (OCR Stage)

Goal: Build the Quantum Neural Network (QNN) for character/layout detection.

- Design "Quantumvolutional" layers to identify document features (edges, text blocks, and noise).
- Train a Variational Quantum Circuit (VQC) for basic character recognition.
- Deliverable: Simulation results of the quantum feature extraction layer.

Week 3: Integration & Raw Extraction

Goal: Connect QML vision output to a digital text stream.

- Interface quantum-processed features with a lightweight classical OCR back-end.
- Refine character extraction accuracy by optimizing quantum circuit parameters.
- Deliverable: Working Python script extracting text from sample documents.

Week 4: Quantum String Operations (Processing Stage) Goal: Implement pattern matching using Grover-based logic.

- Develop a Quantum String Matching algorithm to find specific structural identifiers (e.g., ID numbers).
- Build the oracle for a search complexity of $O(VN)$.
- Deliverable: Functional Grover-based string search simulation.

Week 5: Algorithmic Refinement & Data Cleaning

Goal: Execute usable string operations for data formatting.

· Implement quantum subroutines for string alignment and cleaning of OCR noise.

- Replace LLM-based parsing with targeted quantum logic for specific data schemas.
- Deliverable: Cleaned data output ready for industrial application.

Week 6: Benchmarking & Final Reporting

Goal: Quantitative comparison against classical LLM architectures.

· Compare compute resources (qubit requirements vs. classical GPU/TPU hours).

- Evaluate accuracy vs. model size (Parameters in QNN vs. LLM Weights).
- Deliverable: Final report and presentation of the "Lean Quantum" Document Pipeline.

Tools & Resources

- Coding Tools: Python with Qiskit, PennyLane, and OpenCV.
- Concepts: Grover's Algorithm, Quantum Evolutionary Neural Networks, String Matching

Oracles.

- References: Henderson et al. (Quantum Evolutionary Networks), Grover (Quantum Search).